NEET-SS / FET / INI-SS HIGH-YIELD NOTES

Arthroplasty

1. THA approaches and complications
2. TKA balancing and alignment
3. bearing surfaces and wear
4. periprosthetic infection (ICM criteria)
5. revision arthroplasty

THA approaches and complications

1. Indications and pre-operative planning

- **Indications:** end-stage hip arthritis (primary OA, inflammatory arthritis, AVN, post-traumatic, DDH, Perthes/SCFE sequelae), **displaced femoral neck fractures in active elderly**, failed fixation, tumours.
- **Contraindications:** active infection (joint or systemic), neuropathic (Charcot) joint (relative), severe abductor deficiency (relative: use constrained/dual mobility), medically unfit.
- **Templating** (calibrated AP pelvis, magnification about 20%): **restore the centre of rotation, femoral offset and leg length**; cup at the **teardrop** (medial border) and inferior at the teardrop line; femoral component size and neck cut level; **offset** (reduced offset weakens the abductors and increases impingement/dislocation; excessive offset causes trochanteric pain).
- **Leg length:** measured to the **inter-teardrop line** from the lesser trochanters; **lengthening over about 2-4 cm** risks **sciatic nerve palsy** (peroneal division).

2. Surgical approaches

Approach	Interval / technique	Advantages	Risks
Posterior (Southern/Moore, Kocher-Langenbeck)	Split gluteus maximus; release **short external rotators (piriformis, gemelli, obturator internus) and posterior capsule**	**Commonest**, extensile, preserves abductors, good femoral exposure	**Higher dislocation (posterior)** (reduced by **capsule and rotator repair**, about 1% with repair vs 4% without), **sciatic nerve** injury

Approach	Interval / technique	Advantages	Risks
Direct lateral (Hardinge, transgluteal)	**Split gluteus medius** (anterior third to half) and vastus lateralis (no true internervous plane)	**Low dislocation**, good exposure	**Superior gluteal nerve** injury if the split goes **over about 5 cm above the greater trochanter tip** (abductor weakness, **Trendelenburg gait/limp**), heterotopic ossification
Anterolateral (Watson-Jones)	Between **tensor fasciae latae and gluteus medius** (both superior gluteal nerve: not a true internervous plane)	Low dislocation	Superior gluteal nerve (inferior branch to TFL), abductor damage
Direct anterior (Smith-Petersen/Hueter)	**Superficial: sartorius (femoral) and TFL (superior gluteal)**; **deep: rectus femoris (femoral) and gluteus medius (superior gluteal)**: true internervous plane	Muscle-sparing, **low dislocation**, supine position (fluoroscopy, leg length), possibly faster early recovery	**Lateral femoral cutaneous nerve (meralgia paraesthetica; up to 15-80% transient numbness)**, **femoral fractures** (calcar, greater trochanter) during broaching, wound problems in obese patients, learning curve, **ascending branch of LFCA** (ligate)
Trochanteric osteotomy (Charnley)	Greater trochanter osteotomised	Wide exposure (complex primaries, revision)	**Trochanteric non-union/escape**, wire problems

- **Long-term outcomes are similar** between approaches; choose by surgeon experience and patient factors.
- **Femoral neurovascular structures** lie anterior: anterior retractors placed too medially over the anterior acetabular rim can injure the **femoral nerve/artery**.

3. Acetabular component position

- **Lewinnek "safe zone": inclination (abduction) 40 ±10 degrees**, **anteversion 15 ±10 degrees** (radiographic). Many dislocations occur within this zone, so the **functional (spinopelvic) position** matters: a **stiff lumbar spine** (fusion, degenerative) changes the safe zone (adjust anteversion, consider dual mobility).
- **Combined anteversion (cup + stem) about 25-45 degrees** (Ranawat test/Dorr).
- **Wallace/Hip-spine:** see spinopelvic notes.
- **Steep inclination (over 50 degrees)** = edge loading, increased wear, **squeaking (ceramic-on-ceramic)**, metal ion release (metal-on-metal), dislocation. **Excessive anteversion** = anterior dislocation; **retroversion** = posterior dislocation.
- **Acetabular screws (Wasielewski quadrant system):** a line from the ASIS through the centre of the acetabulum, and a perpendicular: **posterosuperior quadrant = safe** (screws up to about 25 mm+; sciatic nerve and superior gluteal vessels if too long), **posteroinferior = relatively safe** (sciatic, inferior gluteal, internal pudendal), **anterosuperior = "death zone" (external iliac artery and vein)**, **anteroinferior = obturator nerve and vessels**.

4. Fixation

- **Cemented femoral stems: taper-slip (polished, collarless: Exeter, CPT)** subside within the cement mantle creating radial compressive forces (excellent long-term survival), or **composite-beam (shape-closed, matt, collared: Charnley)** bonded to cement. **Third-generation cementing: pulsatile lavage, distal cement restrictor/plug, vacuum mixing (reduces porosity), retrograde filling with a gun, pressurisation, stem**

centralisers, mantle **2 mm or more** (avoid thin mantle/defects: Barrack grading A-D).

- **Uncemented stems:** fixation by **bone ingrowth (porous coating, pore size 50-400 microns)** or **ongrowth (grit-blasted)**; **hydroxyapatite coating** enhances osseointegration; **micromotion under about 40-150 microns** required. **Stress shielding:** proximal bone loss with **stiff, large, extensively coated stems** (Engh); proximal femoral bone loss (Gruen zones 1 and 7).
- **Cups:** mostly uncemented (press-fit, **under-reaming 1-2 mm**, ± screws); cemented polyethylene cups in elderly/poor bone.
- **Age-related trends:** cemented stems reduce **periprosthetic fracture** risk in the elderly (registries: **uncemented stems have higher early periprosthetic fracture rates over about 75 years**).
- **Gruen zones (femur):** 7 zones on AP (1 greater trochanter, 4 tip, 7 calcar), 14 including lateral; **DeLee-Charnley zones (acetabulum)** I-III.

5. Complications

Dislocation

- **About 1-3% after primary THA**; the commonest cause of early revision in some registries.
- **Risk factors: posterior approach without repair**, **component malposition**, **small head size** (**larger heads (32-36 mm) increase jump distance and impingement-free arc**), **reduced offset/soft tissue tension**, abductor deficiency, **neuromuscular disease, cognitive impairment, alcohol**, **spinal fusion/stiffness**, **revision surgery**, female sex, **previous hip surgery**, **fracture as indication**.
- **Early dislocations** (within 3 months) with well-positioned components: **closed reduction** and precautions/brace (about 2/3 do not recur). **Recurrent or malpositioned:** revise: reposition components, **dual-mobility** cup, **larger head**, **constrained liner** (last resort, higher failure), **trochanteric advancement**, abductor reconstruction.
- **Dual-mobility bearings:** small head within a large mobile polyethylene liner inside a polished shell: large effective head, very low dislocation; **intraprosthetic dislocation** is a unique complication.

Periprosthetic fracture (Vancouver classification, femur)

Type	Location	Stem	Treatment
A_G / A_L	Greater / lesser trochanter	Stable	Non-operative unless displaced (A_G: fix if displaced over 2.5 cm/abductor function)
B1	Around/just below the stem	**Stable (well-fixed)**	**ORIF** (locking plate + cables, ± strut allografts)
B2	Around the stem	**Loose**, good bone stock	**Revision with a long stem** (distal fixation, e.g. tapered fluted modular) bypassing the fracture by 2 cortical diameters
B3	Around the stem	**Loose, poor bone stock**	**Revision with proximal femoral replacement** (elderly) or allograft-prosthesis composite (young)
C	Well below the stem tip	Stable	**ORIF** (overlap the stem to avoid stress risers)

- **Distinguishing B1 from B2 is critical**: **a significant proportion of "B1" are actually loose**: test stability intraoperatively; look for radiolucent lines, subsidence, cement mantle fracture.
- **Intraoperative fractures:** commoner with **uncemented press-fit stems**; calcar cracks: **cerclage cable**.
- **Unified Classification System (UCS)** applies to all periprosthetic fractures (see paediatric/trauma notes for type definitions: A apophyseal, B bed of implant (B1 stable, B2 loose, B3 poor bone), C clear of implant, D dividing two implants, E each of two bones, F facing an implant).
- **Acetabular periprosthetic fractures (Paprosky).**

Leg-length discrepancy

- Commonest cause of litigation; lengthening under 1 cm usually tolerated. **True LLD** (bony) vs **apparent/functional** (pelvic obliquity, contractures). **Sciatic palsy risk with lengthening over about 2.5-4 cm.**

Nerve injury (about 0.5-3%)

- **Sciatic nerve (peroneal division)** commonest (posterior approach, lengthening, DDH/revision, retractor placement, cement); **femoral nerve**

(anterior retractors), **superior gluteal nerve** (lateral approach), **lateral femoral cutaneous** (anterior approach), obturator (screws/cement).

Heterotopic ossification (HO)

- **Brooker classification: I** bone islands in soft tissue; **II** spurs with **over 1 cm** between opposing surfaces; **III** under 1 cm gap; **IV** apparent **bony ankylosis**.
- **Risk factors: ankylosing spondylitis, DISH, previous HO**, hypertrophic OA, Paget disease, **male sex**, **lateral/transtrochanteric approaches**, head injury.
- **Prophylaxis** (high-risk patients): **NSAIDs (indomethacin for about 6 weeks)** or **single-dose radiotherapy (about 7-8 Gy) within 24 hours before or 72 hours after surgery**. Excision of mature HO (normal alkaline phosphatase, bone scan quiescent, usually after 6-12 months) with prophylaxis.

Venous thromboembolism

- **DVT without prophylaxis up to about 40-60%**; fatal PE about 0.1-0.5%. **Prophylaxis: LMWH, rivaroxaban/apixaban/dabigatran (DOACs), or aspirin** (**CRISTAL trial** (2022): aspirin had more symptomatic VTE than enoxaparin after arthroplasty; **PEPPER trial**: aspirin, warfarin, rivaroxaban compared), plus **mechanical prophylaxis and early mobilisation**. Extended prophylaxis about **28-35 days after THA** (about 10-14 days after TKA).

Others

- **Infection (about 1%)**, see PJI notes. **Aseptic loosening** and **osteolysis** (wear particles: see bearing notes), **trunnionosis/taper corrosion** (head-neck junction; metal ions; ALTR), **squeaking** (CoC), **iliopsoas impingement** (cup prominent anteriorly; groin pain on resisted hip flexion; **iliopsoas tenotomy or cup revision**), **greater trochanteric pain syndrome**, **vascular injury**, **bone cement implantation syndrome** (hypotension, hypoxia, cardiac arrest during cementing, especially elderly with fractures), **medical complications**.

6. Special situations

- **DDH (Crowe classification by proximal migration of the head): I under 50%**, **II 50-75%**, **III 75-100%**, **IV over 100%** subluxation (relative to femoral head height). **Hartofilakidis** (dysplasia, low dislocation, high dislocation). Place the **cup at the true acetabulum** (anatomical centre), small cups, structural autograft/medialisation, **subtrochanteric shortening osteotomy** (Crowe IV) to prevent sciatic stretch; femur often **narrow and excessively anteverted** (modular or cemented stems to control version).
- **Protrusio acetabuli** (rheumatoid arthritis, Paget, Marfan, osteomalacia): **medial wall bone grafting**, lateralise the centre of rotation, cages/rings in severe cases.
- **Paget disease:** bleeding (consider pre-op bisphosphonates), deformity (osteotomy), hard sclerotic bone, HO.
- **Sickle cell, post-fixation, post-acetabular fracture, ankylosed hip** (abductor function uncertain), **neuromuscular hips**, **young patients** (bearing choice).

Exam pearl: **Posterior approach = commonest, higher dislocation without capsular repair; Hardinge = superior gluteal nerve 5 cm above the trochanter (Trendelenburg); direct anterior = lateral femoral cutaneous nerve and femoral fractures.**

Exam pearl: **Lewinnek: inclination 40 ±10, anteversion 15 ±10. Anterosuperior acetabular quadrant = external iliac vessels (death zone); posterosuperior = safe.** Larger heads and dual mobility reduce dislocation.

Exam pearl: **Vancouver B1 stable = ORIF; B2 loose = long-stem revision; B3 loose + poor bone = proximal femoral replacement/APC; C = ORIF. Brooker HO I-IV; prevent with indomethacin or radiotherapy (7-8 Gy) in high-risk patients.**

TKA balancing and alignment

1. Alignment concepts

- **Mechanical axis of the lower limb:** centre of the femoral head to the centre of the ankle; normally passes **just medial to the centre of the knee** (about 0-3 degrees varus overall).
- **Anatomical vs mechanical femoral axis:** differ by about **5-7 degrees (valgus)**; the **distal femoral cut is made at about 5-7 degrees valgus to the anatomical axis** (perpendicular to the mechanical axis).
- **Joint line:** the native tibial plateau is in **about 3 degrees varus** relative to the tibial mechanical axis; the distal femur in about 3 degrees valgus relative to the femoral mechanical axis; the **joint line is roughly parallel to the floor in stance** (with feet together).
- **Alignment philosophies:**

Philosophy	Principle	Notes
Mechanical alignment (MA)	Femoral and tibial cuts **perpendicular to their mechanical axes** (neutral HKA, joint line horizontal); balance by **soft-tissue releases**	Traditional gold standard; **neutral axis ±3 degrees** associated with better survival (debated with modern implants)
Anatomical alignment (Hungerford)	Joint line at 3 degrees varus, neutral limb	Historical
Kinematic alignment (KA)	**Resurface the pre-arthritic joint** (restore native joint lines and HKA), **minimal releases**	Implants positioned within the patient's native alignment; outlier limb alignments; satisfaction similar or slightly better in some trials

Philosophy	Principle	Notes
Restricted KA / inverse kinematic / functional alignment	KA within safe limits; robotic gap-based adjustments	**CPAK classification** (Coronal Plane Alignment of the Knee: arithmetic HKA and joint line obliquity; 9 phenotypes) describes constitutional alignment

- **Robotic/navigation** TKA reduces alignment outliers; clinical superiority not yet proven.

2. Bone cuts and component rotation

- **Tibial cut:** perpendicular to the tibial mechanical axis in the coronal plane (MA), **posterior slope about 3-7 degrees** (CR knees more; PS knees less, as excess slope can cause instability in flexion/post impingement). Resect about **8-10 mm from the unaffected (higher) side** (measured resection).
- **Distal femoral cut:** about 5-7 degrees valgus to the anatomical axis (intramedullary guide entry point just medial and anterior to the PCL origin); resection equal to the implant thickness (about 9-10 mm).
- **Femoral component rotation (critical for patellar tracking and flexion gap balance):**
  - **Transepicondylar axis (TEA)** (lateral epicondyle to the **medial epicondylar sulcus**: surgical TEA): **most reliable** rotational landmark.
  - **Whiteside line (AP axis):** from the deepest part of the trochlear groove to the centre of the intercondylar notch; femoral component **perpendicular** to it (unreliable in trochlear dysplasia).
  - **Posterior condylar axis (PCA):** femoral component **externally rotated about 3 degrees** relative to it (because the medial posterior condyle is larger); **beware a hypoplastic lateral condyle in valgus knees** (would internally rotate the component: use TEA/Whiteside).
  - **Gap balancing technique:** rotate the femoral component so the **flexion gap is rectangular** after the tibial cut with balanced ligaments.
- **Tibial component rotation:** aligned to the **medial third of the tibial tubercle** (Akagi line: PCL attachment to the medial border of the patellar tendon at its tibial attachment).

- **Internal rotation of femoral or tibial components** (combined) causes **patellar maltracking, lateral subluxation, anterior knee pain, patellar complications and stiffness** (Berger: combined internal rotation 1-4 degrees = tilt; 3-8 = subluxation; 7-17 = dislocation/failure). Assess with **CT**.
- **Anterior femoral notching** (anterior cortex): increases **supracondylar periprosthetic fracture** risk (about 30-40% reduction in torsional strength).
- **Femoral sizing: anterior referencing** (cuts the posterior condyles by the measured amount; downsizing changes the flexion gap: may loosen flexion) vs **posterior referencing** (restores posterior condylar offset; risk of anterior notching or overstuffing the patellofemoral joint).

3. Gap balancing

- **Extension gap** (distal femoral cut + tibial cut) and **flexion gap** (posterior femoral cut + tibial cut) must be **equal and rectangular**.
- **The tibial cut affects both gaps; the distal femoral cut affects only the extension gap; the posterior femoral cuts/femoral size affect only the flexion gap.**

Problem	Flexion gap	Extension gap	Solution
Tight in both	Tight	Tight	**Cut more tibia**
Loose in both	Loose	Loose	**Thicker polyethylene insert** (or augments)
Tight in extension, OK in flexion	Balanced	Tight	**Release posterior capsule / remove posterior osteophytes**, then **resect more distal femur** (about 2 mm)
Loose in extension, OK in flexion	Balanced	Loose	**Distal femoral augmentation** (add distal femur), thicker insert + smaller/posterior cut (combined)
Tight in flexion, OK in extension	Tight	Balanced	**Downsize the femoral component** (anterior referencing) or **increase tibial slope**; release PCL (CR)

Problem	Flexion gap	Extension gap	Solution
Loose in flexion, OK in extension	Loose	Balanced	**Upsize femur** (posterior augments) or **thicker insert + more distal femoral resection**

- **Joint line:** keep within about **8 mm** of native: **elevation of the joint line** (excess distal femoral resection + thick insert) causes **mid-flexion instability, patella baja (pseudo), patellofemoral problems**.

4. Coronal deformity correction (soft-tissue releases)

Varus knee (commonest)

- **Tight medial structures:** remove **medial osteophytes first** (tibial and femoral), then release **deep MCL (meniscotibial)**, **posteromedial capsule/semimembranosus** (tight in **extension**), **superficial MCL** (subperiosteal from tibia or **pie-crusting**; the **anterior sMCL is tight in flexion**, posterior in extension), **pes anserinus** (rarely). **PCL** release (in CR knees with tight flexion).
- Also: **medial tibial reduction osteotomy** (downsize/lateralise the tibial tray and remove medial bone).
- **Fixed flexion deformity:** remove posterior osteophytes, **posterior capsular release**, gastrocnemius recession; as a last resort extra distal femoral resection.

Valgus knee

- **Tight lateral structures: iliotibial band** (tight in **extension**), **popliteus** (tight in **flexion**), **lateral collateral ligament** (tight in **flexion and extension**), **posterolateral capsule** (extension), lateral head of gastrocnemius, biceps femoris.
- **Order (Ranawat inside-out pie-crusting, Whiteside):** posterolateral capsule, ITB, then popliteus/LCL as needed.
- **Hypoplastic lateral femoral condyle**: do not use the PCA alone for rotation.
- **Peroneal nerve palsy** risk when correcting **severe valgus with flexion contracture** (stretch): **flex the knee post-operatively**, loosen dressings;

explore if no recovery.

- **Ranawat valgus types:** I (correctable, medial soft tissue intact), II (fixed, medial attenuated: may need constraint), III (severe osseous deformity with incompetent medial structures: constrained implant).

5. Constraint and design choices

Design	Mechanism	Indications
Cruciate-retaining (CR)	PCL retained; flatter or ultra-congruent inserts	Intact, functional PCL; less bone resection; **paradoxical anterior femoral sliding**
Posterior-stabilised (PS)	**Cam-post** substitutes the PCL (**femoral rollback**)	**Absent/incompetent PCL**, **patellectomy**, inflammatory arthritis, large deformity, previous HTO (debated); risks: **post wear/fracture, patellar clunk syndrome** (fibrous nodule at the superior pole catching the box), cam jump (flexion instability)
Medial pivot / ultra-congruent	Highly conforming medial compartment	Mimics native kinematics
Varus-valgus constrained (CCK, non-linked)	Tall, wide post in a deep box	**MCL/LCL incompetence**, severe valgus, flexion-extension mismatch, revisions
Rotating-hinge (linked)	Hinged femur-tibia linkage with rotation	**Global ligamentous deficiency**, severe bone loss, distal femoral replacement (tumour/fracture), hyperextension instability, **neuromuscular recurvatum** (polio)

- **Use the least constraint needed** (more constraint = more stress transfer to fixation interfaces = loosening).
- **Mobile-bearing vs fixed-bearing**: no proven difference in survival; spinout risk with mobile bearings.
- **Patellar resurfacing:** reduces **reoperation for anterior knee pain** (meta-analyses, KAT trial: little functional difference); inflammatory arthritis

favours resurfacing. **Patellar thickness should be restored** (avoid overstuffing). **Patellar complications:** maltracking (component internal rotation, lateral placement of femoral/tibial components, medial placement of patellar button is preferred), **fracture** (over-resection, avascular after lateral release), **avascular necrosis**, **patellar clunk**, loosening, tendon rupture.

- **Patellar tracking: "no thumb" test**; lateral retinacular release if needed (protect the **superior lateral genicular artery**).

6. Complications of TKA (non-infective)

- **Stiffness/arthrofibrosis** (flexion under about 90 degrees at 6-12 weeks): **manipulation under anaesthesia (within about 6-12 weeks)**, arthrolysis, revision.
- **Instability: flexion instability** (flexion gap larger than extension; recurrent effusions, PS cam jump), **mid-flexion instability**, **genu recurvatum**, global instability: revise with balanced gaps/increased constraint.
- **Aseptic loosening/osteolysis**, **polyethylene wear**.
- **Periprosthetic fractures: supracondylar femur (Lewis-Rorabeck I undisplaced, II displaced + stable prosthesis (ORIF: locking plate or retrograde nail if open box), III loose prosthesis (revision/DFR))**; **tibial (Felix)**; **patella (Goldberg/Ortiguera)**.
- **Peroneal nerve palsy** (valgus, flexion contracture, epidural anaesthesia), **vascular injury** (popliteal artery behind the posterior capsule: **posterolateral corner, about 9 mm from the posterior tibia in flexion**).
- **Wound complications: use the most lateral previous incision** (lateral skin flap less perfused: blood supply from the medial side), skin bridges at least about 5-6 cm.
- **Extensor mechanism disruption:** patellar tendon rupture (**worst; augment with allograft or synthetic mesh, medial gastrocnemius flap**), quadriceps rupture.
- **VTE**, **persistent pain** (about 10-20% dissatisfied after TKA: highest predictors are pre-operative expectations, pain catastrophising, mild radiographic OA).

- **Unexplained painful TKA** work-up: **exclude infection first (ESR/CRP, aspiration)**, then instability, malrotation (CT), loosening, patellar issues, referred pain (hip, spine), CRPS, neuroma.

Exam pearl: **Tibial cut affects both gaps; distal femoral cut affects extension only; posterior femoral cut/femoral size affects flexion only. Tight both = cut more tibia; loose both = thicker insert.**

Exam pearl: **Femoral rotation: TEA most reliable; 3 degrees external to the posterior condylar axis; Whiteside line perpendicular. Internal rotation of components = patellar maltracking. Tibial rotation: medial third of the tubercle.**

Exam pearl: **Varus: medial osteophytes, deep MCL, posteromedial capsule/semimembranosus, then superficial MCL. Valgus: ITB (extension), popliteus (flexion), LCL (both); peroneal palsy risk with valgus + flexion contracture.** Least constraint necessary; rotating hinge for global ligament loss.

Bearing surfaces and wear

1. Tribology basics

- **Wear:** removal of material from articulating surfaces. **Mechanisms:**
  - **Adhesive** (commonest in metal-on-polyethylene: polymer transfers to the counterface and is pulled off),
  - **Abrasive** (hard asperities or particles plough the softer surface),
  - **Third-body** (trapped particles: cement, bone, metal debris),
  - **Fatigue/delamination** (subsurface cracking; important in **TKA**, low-conformity, oxidised polyethylene),
  - **Corrosive** (fretting at modular junctions).
- **Modes of wear (McKellop): Mode 1** intended bearing surfaces; **Mode 2** a primary surface against a non-intended secondary surface (e.g. head against shell after liner wear-through); **Mode 3** third-body particles between primary surfaces; **Mode 4** two non-intended surfaces rubbing (back-side wear of a liner against the shell, neck against the cup rim, **trunnion corrosion**).
- **Lubrication: fluid-film** (ideal; surfaces fully separated: ceramic-on-ceramic and well-designed metal-on-metal with large heads and low clearance), **boundary** (surfaces in contact, lubricant adsorbed: metal-on-polyethylene), **mixed**. Lambda ratio (fluid-film thickness / combined surface roughness) **over 3 = full fluid-film**.
- **Linear wear** (penetration of the head into the liner, mm/year) vs **volumetric wear** (mm^3/year: **determines particle load and osteolysis**). **Volumetric wear increases with head size** ($V = \pi r^2 w$, Charnley); **linear wear decreases with larger heads**.
- **Osteolysis threshold** for conventional polyethylene: **about 0.1 mm/year linear wear** (Dumbleton); above this, osteolysis is common.

2. Biology of particle disease (osteolysis)

- **Wear particles (0.1-1 micron polyethylene most bioactive)** are phagocytosed by **macrophages**, which release **TNF-alpha, IL-1, IL-6, PGE2**, upregulate **RANKL** on osteoblasts/stromal cells: **osteoclast activation** and

bone resorption. The **effective joint space** (Schmalzried) lets particles migrate along the implant-bone interface.

- **Clinical:** often **silent** until large lesions, loosening or fracture; **serial radiographs**, **CT** for extent (pelvic lesions underestimated on radiographs).
- **Treatment: liner exchange and bone grafting** of lytic lesions with retained well-fixed components (risk of dislocation after isolated liner exchange: use larger heads), or full revision for loose components.

3. Polyethylene

- **UHMWPE (ultra-high molecular weight polyethylene):** semi-crystalline polymer. **Gamma irradiation** (sterilisation) breaks chains forming **free radicals**: in air, these **oxidise** (embrittlement, **subsurface white band**, delamination, worse with **shelf ageing**).
- **Sterilisation:** gamma in **inert gas/vacuum** packaging, **ethylene oxide** or **gas plasma** (no cross-linking, no free radicals).
- **Highly cross-linked polyethylene (HXLPE):** irradiated at **5-10 Mrad** to cross-link chains (**much lower adhesive/abrasive wear**: **wear reduced about 80-90%**, dramatic fall in osteolysis and revision for wear in THA); **post-irradiation thermal treatment** removes free radicals: **remelting (above the melting point, about 137 C: eliminates free radicals but reduces crystallinity, fatigue strength and fracture toughness)** or **annealing (below melting: preserves mechanical properties but leaves some free radicals: in-vivo oxidation)**. **Vitamin E (alpha-tocopherol)-infused/blended HXLPE** quenches free radicals without remelting.
- **Cross-linking reduces fatigue strength and toughness**: concern in **TKA (high contact stress, fatigue wear)** and thin liners, **rim/locking mechanism fractures** with thin liners and large heads or malposition.
- **Polyethylene thickness: at least 6-8 mm** in TKA (thinner = high contact stress, delamination).
- **Compression-moulded (direct)** polyethylene has better wear performance than **ram-extruded + machined** stock in TKA; **Hylamer** (failed); **calcium stearate**-free resins.
- **TKA wear:** fatigue-dominated (**delamination, pitting**) because of **low conformity**, rolling/sliding, high contact stresses; **backside wear** of

modular inserts (micromotion against the tray) contributes to tibial osteolysis.

4. Metal-on-polyethylene (MoP)

- **Cobalt-chromium or ceramic heads on HXLPE**: the commonest and most proven bearing in THA (excellent long-term registry survival).
- **Head size: 32-36 mm** heads with HXLPE balance dislocation risk and wear; very thin liners for large heads risk fracture.

5. Ceramics

- **Alumina (Al_2O_3)** and **zirconia-toughened alumina (BIOLOX delta, ZTA)**: **hardest, most scratch-resistant, very low wear, inert particles, excellent wettability (hydrophilic: fluid-film lubrication)**, **lowest wear of all bearings (ceramic-on-ceramic)**.
- **Problems: brittleness** (fracture: much rarer with delta ceramics, about 0.002-0.02%), **squeaking** (about 1-3%: associated with **cup malposition (edge loading)**, **stripe wear**, impingement), **chipping on insertion** (malseating), **limited neck lengths**, cost.
- **Ceramic head fracture:** revise to a **ceramic head** (not metal: **third-body ceramic particles** cause catastrophic metal head wear) with full synovectomy; use a new taper sleeve if the trunnion is damaged.
- **Ceramic-on-polyethylene:** reduces polyethylene wear vs metal heads (smoother, harder, more wettable), avoids taper corrosion (trunnionosis) of CoCr heads.
- **Oxidised zirconium (Oxinium):** zirconium alloy with a **ceramic oxide surface layer**: low wear against polyethylene with metal toughness; **avoid against a damaged surface**: if the thin layer is breached (e.g. after dislocation and closed reduction over the cup rim), the soft substrate wears rapidly.
- **Old zirconia (yttria-stabilised) heads** suffered **phase transformation** (Prozyr recall 2001).

6. Metal-on-metal (MoM)

- **Cobalt-chromium on cobalt-chromium**; large heads possible (hip resurfacing: **Birmingham Hip Resurfacing**; large-head THA). Low volumetric wear in ideal conditions but **nanometre-sized metal particles** (huge numbers) and **metal ions (cobalt, chromium)**.
- **Adverse reactions to metal debris (ARMD) / adverse local tissue reaction (ALTR): pseudotumours** (solid or cystic masses), **ALVAL** (aseptic lymphocyte-dominated vasculitis-associated lesion: type IV delayed hypersensitivity), **metallosis**, **soft-tissue necrosis** (abductor destruction). **Systemic cobalt toxicity:** cardiomyopathy, neuropathy, hypothyroidism, visual/hearing loss.
- **Risk factors: female sex**, **small component size**, **steep cup inclination (over about 50-55 degrees: edge loading)**, **specific designs (ASR recall 2010)**, large-head MoM THA (trunnion corrosion), bilateral implants, renal impairment.
- **Surveillance (MHRA/FDA): blood cobalt and chromium levels** (concern **over about 7 µg/L (ppb)**; some use 2-4 µg/L for unilateral resurfacing as threshold for closer monitoring), **MARS MRI** (metal artefact reduction sequence) or ultrasound for pseudotumours.
- **Revision** for symptomatic patients, rising ions, enlarging pseudotumours or osteolysis; revise to a **ceramic or polyethylene** bearing. **Outcomes of revision for ALTR are worse** than for other aseptic causes.
- **Hip resurfacing: young, active men with large femoral heads** (best results; **avoid in women, small heads, osteoporosis, large cysts, AVN, DDH**). **Femoral neck fracture** (commonest early failure; **notching** of the neck, varus positioning).
- **Metal hypersensitivity** in TKA (nickel) is debated (rarely a true cause of failure; diagnosis of exclusion).

7. Taper corrosion (trunnionosis)

- **Mechanically assisted crevice corrosion** at the **head-neck (and modular neck-stem) junction**: micromotion disrupts the passive oxide layer: fretting + crevice corrosion: **cobalt release**, **ALTR even with metal-on-polyethylene**.

- **Risk factors: large CoCr heads** (larger moment arm/torque), **long necks/high offset**, **dissimilar metals (CoCr head on titanium stem)**, small/short trunnions, **dual-taper modular necks** (recalled designs), poor taper assembly (blood/fluid on the taper, inadequate impaction).
- **Diagnosis: raised cobalt >> chromium** (Co:Cr ratio over about 3-5 suggests taper corrosion rather than bearing wear), **MARS MRI**.
- **Treatment: revise the head to a ceramic head with a titanium sleeve**, change liner, debride; revise the stem if the trunnion is badly damaged.

8. Choosing a bearing (summary)

Bearing	Wear	Particles	Main concerns	Typical use
MoP (CoCr on HXLPE)	Low (with HXLPE)	Polyethylene (osteolysis)	Taper corrosion with large CoCr heads	**Most patients**
Ceramic-on-HXLPE	Lower	Polyethylene	Cost; ceramic fracture (rare)	Younger/active patients; avoid taper issues
Ceramic-on-ceramic	**Lowest**	Ceramic (inert)	**Squeaking, fracture, chipping**, cup position sensitive	Young active patients
MoM	Low (volumetric) but ions	**Metal nanoparticles/ions**	**ALTR/pseudotumour, systemic cobalt**	Largely abandoned except selected resurfacing

- **Hardness (decreasing):** ceramic > CoCr > stainless steel > titanium > polyethylene. **Titanium is never used as a bearing surface** (poor wear resistance, notch sensitivity).
- **Wettability (hydrophilicity):** ceramics best.

Exam pearl: **Volumetric wear determines osteolysis; larger heads = more volumetric, less linear wear (with conventional PE). Osteolysis threshold about 0.1 mm/year. Particles of 0.1-1 micron activate macrophages, TNF-alpha, RANKL.**

Exam pearl: **HXLPE reduces THA wear about 80-90%; remelting removes free radicals but lowers fatigue strength; annealing keeps strength but leaves free radicals; vitamin E is the alternative.** TKA wear is fatigue/delamination (conformity and thickness at least 6-8 mm).

Exam pearl: **Ceramic-on-ceramic: lowest wear; squeaking and fracture; revise a broken ceramic head with a new ceramic head. MoM: pseudotumour/ALVAL, women, small sizes, steep cups; check cobalt/chromium and MARS MRI. Trunnionosis: cobalt much higher than chromium.**

Periprosthetic infection (ICM criteria)

1. Epidemiology and pathogenesis

- **Periprosthetic joint infection (PJI)** occurs in about **1-2% of primary hip and knee arthroplasties** (higher in revisions, about 3-5%). Among the commonest causes of revision (leading cause of early TKA revision).
- **Biofilm:** bacteria adhere to the implant, produce an **extracellular polymeric matrix (glycocalyx)**; sessile bacteria are **up to 1,000 times more resistant to antibiotics** and protected from host defences. **Biofilm matures within about 3-4 weeks**: the basis for implant retention only in early infections.
- **"Race for the surface"** (Gristina): host cells vs bacteria competing to colonise the implant.
- **Organisms: coagulase-negative staphylococci (S. epidermidis) and S. aureus** (together about 50-60%), streptococci, enterococci, gram-negatives, anaerobes; **Cutibacterium acnes** (shoulder, low-grade), fungi (Candida); **polymicrobial** (about 10-15%); **culture-negative** (about 7-15%: prior antibiotics, fastidious organisms).
- **Routes:** perioperative contamination (most), **haematogenous** (late acute; dental, urinary, skin, endocarditis), contiguous spread.

2. Risk factors

- **Patient: diabetes (HbA1c over about 7.5-8%, perioperative glucose over about 200 mg/dl)**, **obesity (BMI over 40)**, **smoking**, **malnutrition (albumin under 3.5 g/dl, transferrin under 200, lymphocytes under 1,500)**, **rheumatoid arthritis/immunosuppression**, **S. aureus colonisation (nasal carriers)**, chronic kidney/liver disease, **prior joint surgery/infection**, **intra-articular steroid injection within about 3 months**, IV drug use, anaemia, male sex, depression.
- **Surgical:** prolonged operative time, **allogeneic transfusion**, haematoma, persistent wound drainage (over about 5-7 days), revision surgery, concurrent infection elsewhere.

3. Prevention

- **Optimise** glucose, BMI, nutrition, smoking cessation (4-6 weeks), treat dental/urinary infection.
- **MRSA/MSSA screening and decolonisation** (**nasal mupirocin + chlorhexidine body washes**).
- **Prophylactic antibiotics: cefazolin within 60 minutes before incision** (vancomycin within 120 minutes if MRSA-colonised or beta-lactam allergic; clindamycin alternative); **redose** after about 4 hours or with large blood loss (over 1,500 ml); **stop within 24 hours**. Antibiotic-loaded cement in high-risk patients.
- **Skin preparation: chlorhexidine-alcohol** (superior to povidone-iodine in surgical site infection trials), hair clipping (not shaving) just before surgery.
- **Theatre: laminar flow**, restricted traffic, **dilute povidone-iodine lavage** before closure (debated benefit), meticulous haemostasis, tranexamic acid (fewer transfusions).

4. Classification by timing (management-oriented)

Type	Timing	Features	Typical organism
Early post-operative (acute)	**Within about 4 weeks (up to 3 months in some systems)** of surgery	Wound drainage, erythema, fever, pain	S. aureus, gram-negatives
Acute haematogenous	Any time, **symptoms under about 3 weeks** in a previously well-functioning joint	Sudden pain, effusion, sepsis	S. aureus, streptococci
Chronic (late)	**Over about 4 weeks**/insidious	Persistent pain, early loosening, sinus tract	CoNS, C. acnes
Positive intraoperative cultures	Unexpected positive cultures at revision for "aseptic" failure	**Two or more positive**: treat as PJI (about 6 weeks)	CoNS, C. acnes

- **Tsukayama classification:** I positive intraoperative cultures, II early post-operative, III acute haematogenous, IV late chronic.

5. Diagnosis

Clinical

- **Pain** (commonest symptom), stiffness, effusion, wound problems, **sinus tract** (pathognomonic), fever (often absent in chronic infection), **early loosening** (within 5 years).

2018 ICM (International Consensus Meeting / Parvizi) definition

- **Major criteria (either = infected):**
  - **Two positive cultures of the same organism** (by standard culture methods).
  - **A sinus tract communicating with the joint or visualisation of the prosthesis.**
- **Minor criteria (pre-operative scoring):**

Criterion	Score
Serum CRP over 1 mg/dL (10 mg/L) or **D-dimer over 860 ng/mL**	**2**
Serum ESR over 30 mm/h	**1**
Synovial WBC over 3,000 cells/μL or **positive leukocyte esterase (++)**	**3**
Positive alpha-defensin (signal-to-cutoff ratio over 1)	**3**
Synovial PMN over 80%	**2**
Synovial CRP over 6.9 mg/L	**1**

  - **Total 6 or more = infected; 2-5 = possibly infected** (add intraoperative findings); **0-1 = not infected**.
- **Intraoperative findings** (for inconclusive pre-operative scores): **positive histology (over 5 neutrophils per high-power field in 5 or more fields: 3 points)**, **purulence (3 points)**, **single positive culture (2 points)**; combined score **6 or more = infected**, 4-5 inconclusive, 3 or less not infected.

- **Thresholds differ for the early post-operative period** (within 6 weeks): **CRP over 100 mg/L (knee/hip acute)**, synovial WBC over about **10,000**, PMN over about **90%** (inflammation from surgery raises markers).
- **Inflammatory arthritis, crystal disease, metallosis (ALTR)** can give false-positive synovial counts (manual count in metallosis).
- **EBJIS 2021 definition** (European Bone and Joint Infection Society) uses "infection likely/confirmed" tiers with similar markers.

Practical work-up (AAOS/ICM)

1. **Serum ESR and CRP** in every painful arthroplasty (high sensitivity: if both normal, PJI unlikely).
2. If raised (or high suspicion): **joint aspiration** (stop antibiotics at least **2 weeks** before aspiration) for **synovial WBC, PMN%, culture** (aerobic + anaerobic; **hold cultures 10-14 days** if C. acnes or low-virulence organisms suspected), **leukocyte esterase**, **alpha-defensin**.
3. **Intraoperative: 5 (3-6) separate tissue samples** with separate instruments, **frozen section**, **sonication** of explants (improves culture yield, especially after antibiotics), avoid swabs.
4. **Imaging:** radiographs (early loosening, periosteal reaction, osteolysis); nuclear imaging (labelled leukocyte with marrow scan) when aspiration is inconclusive; MRI with MARS sequences.

6. Treatment

Debridement, antibiotics and implant retention (DAIR)

- **Indications: early post-operative (within about 4 weeks)** or **acute haematogenous (symptoms under 3 weeks)** infection, **well-fixed implants**, **healthy soft tissue (no sinus)**, susceptible organism.
- **Technique:** radical **synovectomy and debridement**, high-volume lavage (dilute povidone-iodine, chlorhexidine), **exchange of all modular parts** (head and liner/polyethylene insert: essential), followed by **pathogen-specific antibiotics** (**rifampicin** added for staphylococci with retained implants: active against biofilm; never as monotherapy; **fluoroquinolones for gram-negatives**) for about **3 months (hip) to 6 months (knee)**.

- **Success about 50-80%**; worse with **S. aureus/MRSA**, delays, high CRP, rheumatoid arthritis, prior DAIR. **Repeated DAIR** has poor results. **KLIC and CRIME80 scores** predict DAIR failure.

One-stage exchange

- Remove all components, radical debridement, re-implant new components (often **antibiotic-loaded cement**), specific antibiotics. Increasingly used for **chronic infection** with **known organism and susceptibilities**, healthy soft tissues, no sinus (debated), immunocompetent host (Frankfurt/ENDO-Klinik practice). Similar control rates to two-stage in selected cases, fewer operations.

Two-stage exchange (traditional gold standard for chronic PJI, especially in North America)

1. **Stage 1:** remove all components and cement, radical debridement, **antibiotic-loaded cement spacer** (static or **articulating**; high-dose antibiotics: e.g. **vancomycin + gentamicin/tobramycin**; heat-stable antibiotics, about **3-4 g of antibiotic per 40 g cement**), followed by **4-6 weeks of IV/oral antibiotics**.
2. **Antibiotic holiday** (about 2 weeks; debated) and assessment of infection control (**ESR/CRP trends, aspiration**: no single test reliably confirms eradication).
3. **Stage 2:** re-implantation (often after **6-12 weeks**), intraoperative cultures/frozen section.
4. **Success about 85-90%**. **Interim mortality and failure to reach stage 2** are significant in elderly patients.
5. **After re-implantation:** a further **3 months of oral antibiotics** reduced failure in one RCT (Frank et al.).

Salvage

- **Resection arthroplasty (Girdlestone for the hip)**, **knee arthrodesis** (intramedullary nail or external fixation), **amputation** (uncontrolled infection, life-threatening sepsis, massive bone/soft-tissue loss), **long-term suppressive antibiotics** (unfit patients or retained infected implants).

Antibiotic principles

- Bactericidal, biofilm-active agents with good bone penetration: **rifampicin (staphylococci)**, **fluoroquinolones (gram-negatives)**; adjust to cultures; **monitor toxicity** (rifampicin interactions, vancomycin nephrotoxicity, linezolid myelosuppression).
- **Oral vs IV: OVIVA trial** (2019): **oral antibiotics were non-inferior to IV** within the first 6 weeks for bone and joint infection (including PJI), with fewer line complications.
- **Duration: DATIPO trial** (2021): **12 weeks of antibiotics** was superior to 6 weeks for PJI (DAIR/one-stage/two-stage) in persistent infection rates.

7. Special situations

- **Acute haematogenous infection with a well-fixed implant:** DAIR quickly (ideally within days of symptom onset).
- **Fungal PJI:** two-stage, antifungal-loaded spacer (amphotericin/voriconazole), prolonged antifungals; poor outcomes.
- **C. acnes** (shoulder): slow growing (hold cultures 14 days), male, young, previous shoulder surgery.
- **Antibiotic prophylaxis for dental procedures** in arthroplasty patients: **not routinely recommended** (AAOS/ADA), except selected immunocompromised patients.

Exam pearl: **2018 ICM major criteria: two positive cultures of the same organism, or a sinus tract. Minor: CRP/D-dimer (2), ESR (1), synovial WBC over 3,000 or LE ++ (3), alpha-defensin (3), PMN over 80% (2), synovial CRP (1); 6 or more = infected.**

Exam pearl: **DAIR (with modular exchange) for early post-operative (within 4 weeks) or acute haematogenous infection (under 3 weeks symptoms) with stable implants; add rifampicin for staphylococci. Chronic infection: one- or two-stage exchange.** OVIVA: oral non-inferior; DATIPO: 12 weeks beats 6.

Exam pearl: **Biofilm matures in about 3-4 weeks; CoNS and S. aureus are commonest; stop antibiotics 2 weeks before aspiration; take 5 tissue samples; sonicate implants.** Cefazolin within 60 minutes of incision; chlorhexidine-alcohol skin prep.

Revision arthroplasty

1. Why arthroplasties fail

- **THA (registries): aseptic loosening/osteolysis** (commonest long-term), **dislocation/instability**, **infection**, **periprosthetic fracture**, adverse reaction to metal debris, leg-length/pain.
- **TKA: infection** (commonest cause of early revision), **aseptic loosening** (commonest long-term), **instability**, **arthrofibrosis**, polyethylene wear (historical), periprosthetic fracture, malalignment/malrotation, extensor mechanism failure, unexplained pain.
- **Time-dependence: early (under 2 years)**: infection, instability, malposition, fracture; **late**: wear, osteolysis, loosening.
- **Golden rule: every painful or failing arthroplasty is infected until proved otherwise**: **ESR/CRP ± aspiration** before any revision.

2. Evaluation before revision

- History (pain pattern: start-up thigh pain suggests femoral stem loosening; groin/buttock pain suggests acetabular loosening), wound history, systemic symptoms.
- **Infection work-up** (ICM criteria).
- **Serial radiographs: radiolucent lines over 2 mm, progressive**, **subsidence**, **component migration**, cement fracture, **pedestal formation** (distal endosteal bone around an uncemented stem tip: loose), **spot welds** (bone ingrowth: stable), osteolysis, polyethylene wear (eccentric head position).
  - **Engh criteria** for uncemented stem fixation (bony ingrowth vs fibrous stable vs unstable).
  - **Harris criteria** for cemented femoral loosening: **definite** (migration of the stem, cement fracture, stem fracture), **probable** (continuous radiolucent line around the entire cement mantle), **possible** (lucency over 50-100% of the interface).
- **CT** (osteolysis, component version, bone stock), **MARS MRI** (ALTR), **bone scan** (loosening is non-specific), **metal ions** (MoM, taper).

- **Implant identification**, operative notes, extraction tools, components and allograft availability, **neurovascular** proximity (intrapelvic cement/components: CT angiography).

3. Bone loss classifications

Acetabulum: Paprosky

Type	Rim	Columns	Migration / landmarks	Reconstruction
I	Intact, minimal bone loss	Intact	No migration	**Hemispherical uncemented cup**
II (A, B, C)	Distorted (IIA superomedial, **IIB superolateral (rim deficient under 1/3)**, IIC medial wall)	**Intact (supportive)**	**Superior migration under 3 cm**; teardrop and ischium intact	**Hemispherical (jumbo) cup ± screws, particulate graft**
IIIA ("up and out")	Superolateral rim deficient (30-60% of circumference)	Partially supportive	**Over 3 cm superior migration**, **Kohler line intact**, moderate ischial and teardrop lysis	**Jumbo cup + porous metal augments (trabecular metal)** or structural allograft
IIIB ("up and in")	Over 60% rim loss	**Non-supportive**	**Kohler line breached**, severe ischial lysis, teardrop obliterated; **pelvic discontinuity** risk	**Cup-cage**, **custom triflange**, **augments + cage**, **distraction technique**

- **Pelvic discontinuity:** separation of the superior and inferior hemipelvis through the acetabulum (**fracture line through both columns, medial translation/rotation of the inferior hemipelvis, broken Kohler line**). Treat with **posterior column plating + cup**, **cup-cage**, **acetabular distraction**, or **custom triflange**.

- **AAOS classification:** I segmental, II cavitary, III combined, IV pelvic discontinuity, V arthrodesis.

Femur: Paprosky

Type	Bone loss	Reconstruction
I	Minimal metaphyseal loss (like primary)	Primary-type stem (cemented or uncemented)
II	**Metaphyseal loss**, **intact diaphysis**	**Extensively porous-coated (cylindrical) stem** or **modular tapered stem**
IIIA	Metaphysis non-supportive, **over 4 cm of intact isthmus** for scratch fit	**Extensively porous-coated stem** (diaphyseal fixation) or **modular tapered fluted (Wagner-type)**
IIIB	**Under 4 cm** of intact isthmus	**Modular tapered fluted titanium stem**
IV	**Extensive diaphyseal loss, widened canal**, no isthmus support	**Impaction allografting (cemented, Exeter technique)**, **allograft-prosthesis composite (young)**, **proximal femoral replacement (elderly)**

- **Extensively coated cylindrical cobalt-chromium stems** fail when the canal is **over about 18-19 mm** (stiff, stress shielding, less isthmus); **tapered fluted titanium stems** (modular or monoblock) are now widely used for most femoral revisions (reliable distal fixation, rotational stability via flutes).
- **Extended trochanteric osteotomy (ETO):** for **removing well-fixed stems/cement**, correcting proximal deformity (varus remodelling); length about **10-15 cm** (two-thirds of the trochanter with the lateral cortex); fix with cables; allows distal fixation **at least 4-6 cm below the osteotomy**.

Knee: Anderson Orthopaedic Research Institute (AORI)

Type	Bone loss	Reconstruction
1	**Intact metaphysis**, small defects	**Cement ± screws**, standard components
2A	Damaged metaphysis, **one condyle/plateau**	**Augments (metal wedges/blocks)**, particulate graft

Type	Bone loss	Reconstruction
2B	Damaged metaphysis, **both condyles/plateaus**	Augments, **cones/sleeves**
3	**Deficient metaphysis**, often with collateral ligament/tendon detachment	**Metaphyseal cones (porous tantalum/titanium) or sleeves**, **structural allograft**, **hinged prosthesis**, **distal femoral/proximal tibial replacement**

- **Three-zone fixation (Morgan-Jones): epiphysis (zone 1), metaphysis (zone 2), diaphysis (zone 3)**: aim for **solid fixation in at least two zones**; **metaphyseal fixation (cones/sleeves)** is key.
- **Stems: cemented** (shorter, fills canal) or **press-fit diaphyseal-engaging** (offset couplers to avoid tibial/femoral conflict).

4. Revision TKA principles

1. **Exposure:** medial parapatellar arthrotomy, synovectomy, release scar in the gutters, **external rotation of the tibia and medial release**; for difficult exposure: **quadriceps snip** (first choice; proximal oblique incision of the quadriceps tendon), **V-Y quadricepsplasty (turndown)** (extensor lag), **tibial tubercle osteotomy (TTO)** (long osteotomy, about 6-8 cm, fixed with wires/screws). **Avoid patellar tendon avulsion** (catastrophic): pin the tendon, use the snip/TTO early.
2. **Component removal:** preserve bone (osteotomes, saws, ultrasonic tools).
3. **Restore the tibial platform** first (establishes the joint line), then the **femur in flexion (size, rotation, posterior augments)**, then **extension**.
4. **Joint line restoration:** about **2.5 cm (one fingerbreadth) above the fibular head**, about **3 cm distal to the medial epicondyle**, **1 cm below the inferior pole of the patella**; avoid elevation (patella baja, mid-flexion instability).
5. **Constraint as needed** (PS, VVC, rotating hinge) to achieve stability with balanced gaps.
6. **Patella:** revise if loose/worn; patellar bone loss: **gull-wing osteotomy**, porous metal patellar augments, patelloplasty.

5. Revision THA principles

- **Approach:** usually posterior/extensile; **ETO** for stem/cement removal.
- **Acetabulum: uncemented hemispherical cups with screws** handle most defects (Paprosky I-IIIA), with **porous metal augments** for segmental loss; **impaction grafting with cemented cup** (Slooff) for contained defects; **cages/cup-cage/triflange** for IIIB/discontinuity.
- **Femur:** as per Paprosky; **cement-in-cement revision** (well-fixed intact cement mantle: re-cement a new taper-slip stem into the old mantle) for rapid revision with retained cement.
- **Instability prevention in revision: larger heads, dual-mobility cups** (reduce dislocation markedly in revision), **constrained liners** for abductor deficiency (high failure), **abductor reconstruction**.
- **Periprosthetic fracture revision** (Vancouver B2/B3) as in THA notes.

6. Complications of revision

- **Infection (higher than primary)**, **dislocation (up to about 10-15% in revision THA)**, **intraoperative fracture/perforation**, nerve injury, **aseptic re-loosening**, **extensor mechanism failure (TKA)**, wound breakdown (use previous lateral incision; **plastic surgery** input; medial gastrocnemius flap for exposed TKA), **thromboembolism**, **blood loss** (tranexamic acid, cell saver), **medical complications** (elderly).

7. Megaprostheses and salvage

- **Proximal femoral replacement (PFR)** / **distal femoral replacement (DFR)** / **total femur**: tumour resections, massive bone loss, elderly periprosthetic fractures (allows immediate weight bearing). **High dislocation (PFR)** and **infection rates (about 10%+)**; use **silver-coated** implants, mesh for soft-tissue attachment.
- **Allograft-prosthesis composite (APC):** younger patients (biological soft-tissue attachment).
- **Knee arthrodesis** (failed multiple revisions/infection with extensor loss), **above-knee amputation**, **resection arthroplasty** (Girdlestone hip).

Exam pearl: **Every failing arthroplasty is infected until proved otherwise. TKA: infection = commonest early failure; THA long-term: aseptic loosening/osteolysis.**

Exam pearl: **Paprosky acetabulum: II (rim/columns supportive, under 3 cm migration) = hemispherical cup; IIIA (up and out, Kohler intact) = jumbo cup + augments; IIIB (up and in, Kohler broken) = cup-cage/triflange; watch for pelvic discontinuity.**

Exam pearl: **Paprosky femur: II/IIIA = extensively porous-coated or modular tapered; IIIB (under 4 cm isthmus) = modular tapered fluted; IV = impaction grafting/APC/PFR. AORI knee: 1 cement, 2 augments/cones, 3 cones/sleeves/hinge/replacement; fix in two of three zones. Revision TKA exposure: quadriceps snip first; protect the patellar tendon.**